THE PROJECTION FORMULA IN THE DERIVED CATEGORY FOR A FINITE MORPHISM OF ALGEBRAIC SPACES WITH SHEAVES OF RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Exactness of the pushforward along a finite morphism. Same chapter spaces-cohomology.tex, label lemma-finite-higher-direct-image-zero (invoked at proof line 622). Because f_* is exact, Rf_* is computed by applying f_* to any representative complex, which turns the derived statement into a termwise one.
- K-flat resolutions and their behaviour under pullback. Cohomology on Sites, labels sites-cohomology-lemma-K-flat-resolution (invoked at proof line 626) and sites-cohomology-lemma-pullback-K-flat (invoked at proof line 628). They provide a K-flat complex with flat terms representing K, whose pullback is again K-flat with flat terms, so that the derived tensor products are computed by the naive total complexes.
- The pushforward along a finite morphism commutes with direct sums, and the non-derived projection formula. Same chapter spaces-cohomology.tex, labels lemma-stalk-push-finite (invoked at proof line 641) and lemma-finite-rings (invoked at proof line 648). The first, through the stalk description, gives the commutation with direct sums used to pass to terms; the second is exactly the termwise equality $\mathcal{K}^n \otimes_\mathcal{B} f_*\mathcal{M}^m = f_*(f^*\mathcal{K}^n \otimes_\mathcal{A} \mathcal{M}^m)$.

Source reference. Original lemma 0A4K: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 496–496. Navigation pointer for \ref{lemma-finite-higher-direct-image-zero} used inside the counted span (line 622); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DK4: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 543–543. Navigation pointer for \ref{lemma-finite-rings} used inside the counted span (line 612); the argument that consumes it is retained in the span.

Source reference. Original lemma 0DK3: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 518–518. Navigation pointer for \ref{lemma-stalk-push-finite} used inside the counted span (line 641); the argument that consumes it is retained in the span.

Source reference. Original lemma 06YS: sources/stacks-chow/tags.txt, source lines 8896–8896. Navigation pointer for \ref{sites-cohomology-lemma-K-flat-resolution} used inside the counted span (line 626); the argument that consumes it is retained in the span.

Source reference. Original lemma 0G7E: sources/stacks-chow/tags.txt, source lines 20229–20229. Navigation pointer for \ref{sites-cohomology-lemma-pullback-K-flat} used inside the counted span (line 628); the argument that consumes it is retained in the span.

Lemma 0.1. *With S, X, Y, π, $\mathcal{A}$, $\mathcal{B}$, φ, and f as in Lemma 0DK4 we have*

$$K \otimes^{\mathbf{L}}_{\mathcal{B}} Rf_*M = Rf_*(Lf^*K \otimes^{\mathbf{L}}_{\mathcal{A}} M)$$

in $D(\mathcal{B})$ for any $K \in D(\mathcal{B})$ and $M \in D(\mathcal{A})$.

Proof. Since f_* is exact (Lemma 0A4K) the functor Rf_* is computed by applying f_* to any representative complex. Choose a complex $\mathcal{K}^\bullet$ of $\mathcal{B}$-modules representing K which is K-flat with flat terms, see Cohomology on Sites, Lemma 06YS. Then $f^*\mathcal{K}^\bullet$ is K-flat with flat terms, see Cohomology on Sites, Lemma 0G7E. Choose any complex $\mathcal{M}^\bullet$ of $\mathcal{A}$-modules representing M. Then we have to show

$$\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{B}} f_*\mathcal{M}^\bullet) = f_*\mathrm{Tot}(f^*\mathcal{K}^\bullet \otimes_{\mathcal{A}} \mathcal{M}^\bullet)$$

because by our choices these complexes represent the right and left hand side of the formula in the lemma. Since f_* commutes with direct sums (for example by the description of the stalks in Lemma 0DK3), this reduces to the equalities

$$\mathcal{K}^n \otimes_{\mathcal{B}} f_*\mathcal{M}^m = f_*(f^*\mathcal{K}^n \otimes_{\mathcal{A}} \mathcal{M}^m)$$

which are true by Lemma 0DK4. □